Mongol Empire Quantitative Data Appendix (c. 1206–1368)

CAS Framework Empire Knowledge Base — Session 4 of 4 (FINAL)

The Mongol Empire represents the largest contiguous land empire in human history, Encyclopedia Britannica spanning Encyclopedia Britannica **24 million km²** at its peak and ruling approximately **100 million people**—roughly one-quarter of the world's population. This appendix compiles quantitative data across economic, demographic, financial, institutional, and military domains, marking the completion of the Infonomics Project's seven-empire comparative framework. The Mongol case presents unique methodological challenges: a nomadic economy that defies traditional GDP metrics, primary sources predominantly in Chinese, Persian, and European languages rather than Mongol, and conquest mortality estimates that remain among the most contested figures in medieval historiography.

Unlike the sedentary empires in this series, Mongol wealth flowed through tribute extraction, trade facilitation, and conquest plunder rather than agricultural taxation. The Yuan dynasty's sophisticated paper money system provides the empire's most quantifiable financial data, while the yam postal network—exceeding **1,400 stations** with **50,000 horses**— Wikipedia offers concrete institutional metrics. Wikipedia Military data relies heavily on Carl Sverdrup's recent revisionist scholarship, which has dramatically revised traditional army sizes downward. Throughout this appendix, confidence levels reflect the inherent limitations of medieval sources: chronicler casualty figures for sieges like Baghdad and Merv are treated as indicators of catastrophic events rather than reliable counts.

---

1. Economic indicators

1.1 Tribute and tax revenue systems

The Mongol fiscal apparatus combined steppe traditions with conquered administrative systems, creating a hybrid structure that varied significantly across the empire's four successor khanates.

Table 1.1: Core Tax Categories

Tax Type	Rate/Structure	Geographic Application	Source	Confidence
Tamga (commercial tax)	3–5% on trade turnover	Empire-wide, especially Silk Road cities	Poletaev; CyberLeninka	MODERATE
Tamga (capital-based)	1 dinar per 240 dinars (~0.42%)	Persian/Central Asian merchants	GlobalDomainNews	LOW
Qubchur (poll tax)	Flat-rate or graduated levy	Semi-nomadic and settled subjects	UNESCO Silk Road	HIGH
Alba (livestock tax)	1 sheep per flock (originally)	Mongol pastoral territories	Ögedei decree 1229; UNESCO	HIGH

The tamga commercial tax, etymologically derived from the Turkic word for "stamp" or "seal," became so embedded in Eurasian commerce that it later formed the basis of Russia's customs system (tamozhnia). First decreed imperially in **1234–1235** under Ögedei Khan, the tamga generated **1,582,560 silver taels** in Yuan China by 1328—representing **5.71%** of total central government revenue. Pku

Table 1.2: Russian Principalities' Tribute to Golden Horde

Principality	Annual Amount	Period	Source	Confidence
Total Russian principalities	12,000–14,000 rubles (~1.5 tons silver)	Mid-14th century	GlobalDomainNews	MODERATE
Vladimir principality	5,000 rubles/year	Mid-14th century	GlobalDomainNews	MODERATE
Novgorod lands	2,000 rubles/year	Mid-14th century	GlobalDomainNews	MODERATE
Tver lands	2,000 rubles/year	Mid-14th century	GlobalDomainNews	MODERATE
Moscow principality	1,280 rubles/year	Mid-14th century	GlobalDomainNews	MODERATE
Total (declining phase)	5,000–7,000 rubles/year	1380–1472	Wikipedia	MODERATE
Final phase	~1,000 rubles/year	Post-1472	Wikipedia	MODERATE

Tribute amounts fluctuated dramatically with Golden Horde power. The basic rate during early occupation approximated one-tenth of family income—a "tithe" structure. By 1275, North-East Russia paid **1/2 grivna silver per sokha** (plow/farm unit). Per-family silver burden estimates range from **75–100 grams annually**, though enforcement varied considerably.

1.2 The Ortaq merchant partnership system

The ortaq (also ortogh, from Turkic "partner") represents one of history's earliest sophisticated merchant-state partnership structures, resembling Islamic qirad, Italian commenda, and mudharaba arrangements. Taylor & Francis Online Taylor & Francis Online

Table 1.3: Ortaq System Structure

Characteristic	Description	Source	Confidence
Institutional establishment	General Administration for Supervision of Ortogh created 1268	Cambridge History of China	HIGH
Capital providers	Mongol state, aristocrats, imperial family	Wikipedia; Enkhbold (2019)	HIGH
Operations managers	Primarily Uighur and Muslim merchants	Columbia AFE	HIGH
Liability structure	Principal liable only to original investment	Enkhbold (2019); Richmond Fed	HIGH
Capital forms	Metal coins, paper money, gold/silver ingots, tradable goods	Enkhbold (2019)	HIGH
Typical caravan size	70–100 men per mission	Columbia AFE	MODERATE
Trade privileges	Use of official yam relay stations	Wikipedia	HIGH
Special privileges	Right to bear arms	Endicott-West (1989)	HIGH

The ortaq system channeled state capital through merchant expertise, creating a risk-pooling mechanism that enabled long-distance Silk Road commerce. Marco Polo's family participated in documented ortaq partnerships. However, ortaq merchants acquired a "low reputation among Chinese for moneylending at high interest"—the Richmond Federal Reserve notes they charged **100% compound annual interest rates**, "unencumbered by usury laws of Islamic and Christian worlds."

Data Gap: Specific profit-sharing percentages between state capital providers and merchant operators remain undocumented in surviving sources.

1.3 Trade route data during the Pax Mongolica

The Pax Mongolica (c. 1250–1350) enabled unprecedented Eurasian commercial integration, National Geographic though quantified trade volumes remain elusive.

Table 1.4: Silk Road Infrastructure (Yuan China)

Component	Quantity	Source	Confidence
Postal/relay stations	>1,400	Marco Polo; Wikipedia	HIGH
Horses maintained	50,000	Marco Polo; Lumen Learning	MODERATE
Oxen	8,400	Lumen Learning	MODERATE
Mules	6,700	Lumen Learning	MODERATE
Carts	4,000	Lumen Learning	MODERATE
Boats	6,000	Lumen Learning	MODERATE

Janet Abu-Lughod's seminal *Before European Hegemony* (1989) Wikipedia identifies eight interconnected trade circuits operating during the Mongol period, with no single hegemon dominating the system: Goodreads

Table 1.5: Abu-Lughod's Eight Trade Circuits (c. 1250–1350)

Circuit	Geographic Region	Key Nodes
I	Champagne Fairs/Northern Europe	Bruges, Troyes, Provins, Ghent
II	Mediterranean (Italian)	Genoa, Venice
III	Eastern Mediterranean/Levant	Constantinople, Cairo
IV	Middle East/Baghdad	Baghdad, Hormuz
V	Arabian Sea/Red Sea	Aden
VI	Indian Subcontinent	Gujarat, Malabar, Calicut
VII	Straits of Malacca	Malacca, Palembang
VIII	East/Southeast Asia	Hangzhou (Kinsay), Canton, Quanzhou

Silver flow proxies from the Tilburg University thesis (de Graaf) identify peak trade periods at **1276–1288**, **1299–1309**, and **1344–1359**. Uvt The system collapsed due to the Black Death (~1350) and Ming isolation policies. Open Library

Critical Data Gap: No quantified trade volume estimates in tonnage or monetary value exist for Silk Road commerce during this period. All assessments remain qualitative ("flourishing," "extensive"). A single documented caravan included 450 Muslim traders with 500 camels, but such examples cannot be aggregated.

1.4 Economic output estimates

The most reliable quantitative data comes from Yuan dynasty fiscal records, particularly the comprehensive 1328 revenue breakdown reconstructed by Guan, Ma, and Zhai (2024) from Peking University.

Table 1.6: Yuan Dynasty Central Government Revenue (1328)

Revenue Category	Amount (silver taels)	Share (%)	Confidence
Total Revenue	**27,736,844**	**100%**	HIGH
Salt tax	12,768,333	**46.03%**	HIGH
Shuiliang (land tax)	8,325,260	30.02%	HIGH
Seigniorage (paper money)	2,562,595	9.24%	HIGH
Commercial tax (tamga)	1,582,560	5.71%	HIGH
Wine	782,603	2.82%	HIGH
Kechai (poll tax)	568,048	2.05%	HIGH
Tea	482,018	1.74%	HIGH
Gold	244,707	0.88%	HIGH
Miscellaneous	278,014	1.00%	HIGH
Silver	77,561	0.28%	HIGH
Iron	21,561	0.08%	HIGH
Vinegar	37,660	0.14%	HIGH

The salt monopoly's dominance (**46%** of revenue, sometimes exceeding **60–80%** in other years) reveals a distinctively "modern" fiscal structure. Yuan China displayed the **lowest dependence on agricultural taxation** of any Chinese dynasty—just 30% versus 86–91% for the Ming and 73–80% for the Qing. pku Pku This commercial orientation made the Mongol-ruled economy structurally distinct from its predecessors and successors.

Table 1.7: Comparative Chinese Dynasty Revenues

Dynasty	Total Revenue (silver taels)	Agricultural Tax Share	Source
Northern Song (1085)	~60 million	~51%	Guan et al. (2024)
Yuan (1328)	~27.7 million	~30%	Guan et al. (2024)
Ming (various)	<50 million	~86–91%	Guan et al. (2024)
Qing (early)	~30–50 million	~73–80%	Guan et al. (2024)

Maddison Project GDP per capita estimates for China c. 1300 suggest approximately **$600 in 1990 International Dollars**, though this figure is heavily contested. Angus Maddison himself argued Chinese economy "suffered setbacks under the Yuan" followed by stasis from c. 1300–1850, but scholars including Bairoch, Pomeranz, Frank, and Denemark have challenged his methodology for potentially underestimating Asian per-capita income.

Data Gap: GDP estimates for non-Yuan khanates (Ilkhanate, Golden Horde, Chagatai) are essentially unavailable. The nomadic core economy resists traditional GDP metrics entirely.

---

2. Demographic indicators

Population data for the Mongol Empire involves some of medieval history's most contested figures. The challenge is twofold: pre-conquest baselines are poorly documented, and post-conquest censuses were demonstrably incomplete. Modern scholars increasingly question the extreme mortality estimates that dominated earlier historiography.

2.1 Population of conquered territories

Table 2.1: Chinese Population Estimates

Territory/Period	Population Estimate	Source	Confidence
Jin Dynasty (Northern China) 1207 census	53.5 million (8.4 million households)	Jin Shi; Kuhn reconstruction	MODERATE
Southern Song (1207)	75–80 million (13.6 million households)	Song Shi	MODERATE
Western Xia (Tangut)	~3 million	John Man	LOW
Yuan China (post-conquest census)	58–60 million registered	Yuan Shi	LOW-MODERATE
Ming founding (1368)	~60 million	Contemporary estimates	MODERATE
Ming (1393 census)	65 million	Census records	MODERATE

John Durand's *Population Statistics of China, A.D. 2–1953* casts significant doubt on dramatic population decline narratives. Wikipedia Yuan censuses "likely missed large portions of population due to incomplete registration." Wikipedia Diana Lary emphasizes massive **southward migration** rather than pure mortality—population displacement on "a scale never seen before." Wikipedia

Table 2.2: Persian and Central Asian Population Estimates

Region	Pre-Conquest	Post-Conquest	Source	Confidence
Khwarazmian Empire	~5 million	Unknown	Chandler & Fox	MODERATE
Greater Iran (deaths attributed)	—	1–3 million direct; 2–7 million total	Modern scholarly synthesis	MODERATE
Persian (extreme claim)	2.5 million	250,000	Smith	VERY LOW

The Persian population figures attributed to Smith represent an extreme position now considered exaggerated. Modern scholarly consensus suggests **1–3 million direct deaths** in Greater Iran with **2–7 million total** including famine and displacement mortality. Quora

Table 2.3: Russian Principalities

Metric	Value	Source	Confidence
European Russia pre-invasion	7.5 million	Colin McEvedy	MODERATE
European Russia post-invasion	7 million	Colin McEvedy	MODERATE
Net decline	~500,000	McEvedy	MODERATE

McEvedy's estimate of relatively modest Russian decline (~500,000 or 6.7%) contrasts sharply with popular narratives of devastation. Wikipedia

2.2 City populations before and after conquest

Table 2.4: Major City Populations and Siege Impacts

City	Date	Pre-Siege Population	Deaths Reported (Medieval)	Deaths (Modern Estimates)	Confidence
Baghdad	1258	500,000–1 million	800,000–2 million	90,000–200,000	MODERATE (modern)
Kiev	1240	40,000–50,000	Near-total	~2,000 survivors	MODERATE
Merv	1221	<70,000 actual	700,000–1.3 million	Unquantifiable	LOW
Nishapur	1221	100,000–200,000	1–1.7 million	Unquantifiable	LOW
Samarkand	1220	<70,000	30,000	—	MODERATE
Hangzhou (pre-1276)	—	1.1 million	—	—	MODERATE
Karakorum (Mongol capital)	Peak	10,000–30,000	—	—	MODERATE

The Baghdad siege of 1258 illustrates the scholarly challenge. **Hulagu Khan's own estimate** was approximately **200,000 deaths**. University of Hawaii The Persian historian Hamdallah Mustawfi (writing in 1330) claimed 800,000. Other chroniclers reached 2 million. Modern scholars like Martin Sicker suggest ~90,000. The city's revenues fell to **one-tenth of pre-Mongol levels** by 1335—quantifiable economic impact even if mortality remains uncertain.

For Central Asian cities, Chandler and Fox established that Merv, Samarkand, and Nishapur were "vying for the largest" with populations **less than 70,000 each**. Medieval chronicler claims of 1.3–2.4 million deaths are therefore logistically impossible and should be treated as literary expressions of catastrophe rather than census data. (Quora)

Giovanni da Pian del Carpine's 1246 eyewitness account of Kiev provides the most reliable siege aftermath data: approximately **2,000 survivors** in a city that previously held 40,000–50,000, with only **~200 houses remaining** of the original urban fabric.

2.3 Crisis mortality estimates: the scholarly debate

Table 2.5: Total Conquest Mortality Estimates

Estimate	Source	Methodology	Confidence
40–60 million	McEvedy & Jones (1978)	Census comparison, extrapolation	LOW
35 million (China alone)	McEvedy & Jones citing Durand	Census decline analysis	LOW
30 million total	R.J. Rummel (Hawaii)	Chronicle compilation	LOW-MODERATE
20–60 million	Wikipedia synthesis	Combined campaigns 1206–1405	LOW
~11.5 million	Recent revisionist studies	Reduced estimates over 120 years	MODERATE

The commonly cited "40 million deaths" originates from McEvedy and Jones's 1978 *Atlas of World Population History*, a source now considered methodologically weak for this specific claim. (Wikipedia) John Durand—whose work McEvedy and Jones cite—himself questioned the conclusions drawn from Yuan census deficiencies. (Wikipedia)

Key Scholarly Positions:

- **John Durand**: Lower estimates; Mongol censuses likely very incomplete, artificially exaggerating apparent decline
- **Diana Lary**: Migration emphasis; southward population displacement rather than purely mortality
- **Colin McEvedy & Richard Jones**: Higher estimates (40–60 million); methodology now heavily criticized
- **R.J. Rummel**: ~30 million total; acknowledges extreme uncertainty
- **Timothy May**: Massacres used "judiciously" as strategic terror, not indiscriminate (Cambridge Core)

2.4 Population recovery trajectories

Table 2.6: Recovery Timelines by Region

Region	Recovery Period	Key Indicators	Confidence
China	~200–300 years (1280→1600)	Population reached 150 million by 1600	MODERATE
Persia/Iran	Never fully recovered until modern era	Merv still in ruins 1330s; Baghdad revenues 1/10th by 1335	MODERATE
Kiev/Rus'	>100 years for significant recovery	Stone construction "halted for a long time"	MODERATE

Persian recovery was particularly impeded by **qanat irrigation system destruction**—permanent agricultural infrastructure loss that affected food production for centuries. Some cities like Merv and Nishapur never recovered their pre-conquest prominence.

2.5 Total population under Mongol rule at peak

Table 2.7: Peak Empire Population (c. 1260–1280)

Metric	Estimate	Source	Confidence
Total population ruled	~100 million	Multiple encyclopedia sources	MODERATE
World population c.1260	~400–450 million	Demographic estimates	MODERATE
Share of global population	~25%	Calculated	MODERATE
Empire territorial extent	24 million km²	Britannica	HIGH
Yuan China alone	58–60 million registered	Yuan Shi census	MODERATE

---

3. Financial indicators

3.1 Yuan dynasty paper money system

The Yuan jiaochao represents history's first large-scale paper currency experiment and provides the Mongol Empire's most quantifiable financial data.

Table 3.1: Paper Money Issues (Chronological)

Currency Name	Issue Date	Circulation Period	Value Relationship	Source	Confidence
Zhongtongchao	1260	1260–1286	Base currency	Guan, Palma & Wu (2024)	HIGH
First Zhiyuanchao	1287	1287–1309	1 guan = 5 guan zhongtongchao	Guan, Palma & Wu (2024)	HIGH
Zhidachao	1310	1310 only (1 year)	1 liang = 25 guan zhongtongchao	Guan, Palma & Wu (2024)	HIGH
Second Zhiyuanchao	—	1311–1351	—	Guan, Palma & Wu (2024)	HIGH
Zhizhengchao	1352	1352–1368	1 guan = 10 guan zhongtongchao	Guan, Palma & Wu (2024)	HIGH

Table 3.2: Denominations and Exchange Rates

Metric	Value	Date	Source	Confidence
Zhongtongchao denominations	10 denominations: 10–2,000 wen	1260	Yuanshi	HIGH
Paper-to-silver rate	2 guan = 1 liang silver	1260	Guan et al. (2024)	HIGH
Paper-to-silver rate	10 guan = 1 liang silver	1287	Yuanshi ch. 93	HIGH
Paper-to-silver (official)	25 guan = 1 liang silver	1309	Li (2014)	HIGH
Paper-to-silver (market)	52.7 guan = 1 liang silver	1309	Li (2014)	MODERATE

Table 3.3: Monetary Standard Periods

Period	Dates	Characteristics	Confidence
Full Silver Standard	1260–1275/76	Full silver convertibility at fixed exchange rate	HIGH
Nominal Silver Standard	1276–1309	Silver convertible only when available at exchange bureaus	HIGH
Fiat Standard	1310–1368	Silver no longer played reserve role; de jure fiat money	HIGH

The transition from silver-backed to fiat currency represents a crucial turning point. After 1310, Yuan paper money functioned as pure fiat currency—revolutionary for its era but ultimately unsustainable.

Table 3.4: Inflation Data

Period	Compound Annual Inflation	Source	Confidence
1260–1290	11.0%	Guan, Palma & Wu (2024)	HIGH
1290–1340	1.8%	Guan, Palma & Wu (2024)	HIGH
1340–1355	4.5%	Guan, Palma & Wu (2024)	HIGH
1346–1355	12.7%	Guan, Palma & Wu (2024)	HIGH
Total depreciation (1260–1309)	**~1,000%**	Ancient Origins	MODERATE
Final price level (by 1368)	**140–800x initial**	Yuanshi; Guan et al.	MODERATE

The inflation trajectory shows remarkable stability during **1290–1340** (only 1.8% annual), followed by accelerating crisis after 1340. Peak issuance occurred in **1354 at 60 million ding**—the highest in Yuan history—as the dynasty attempted to fund military operations against rebellions. By 1368, paper money was effectively worthless. Ehs

Table 3.5: Silver Exchange Bureau Network (1290s)

Metric	Value	Source	Confidence
Total exchange bureaus	65	Yuandianzhang	HIGH
Bureaus in capital (Dadu)	6	Yuandianzhang	HIGH
Bureaus in Yangzhou	3	Yuandianzhang	HIGH
Exchange discount	70–80% of face value	von Glahn (1996)	HIGH
Commission for spoiled notes	3%	von Glahn (1996)	HIGH

3.2 Silver and gold standards

Table 3.6: Ingot Currency (Ding/Sycee)

Metric	Value	Source	Confidence
Standard ding weight	50 liang silver	Guan et al.; chinaknowledge.de	HIGH
Liang (tael) weight	~36–37.5 grams	Wikipedia; chinaknowledge.de	HIGH
Standard ding in grams	~1,850 grams (~1.85 kg)	Calculated	HIGH
Paper money equivalent	1 ding = 50 guan zhongtongchao	Guan et al. (2024)	HIGH

Table 3.7: Gold-Silver Ratios (Comparative)

Region/Period	Ratio	Source	Confidence
China (early period)	1:3 or lower	Dokumen.pub	MODERATE
China (early Ming, ~1400)	1:5 to 1:6	Dokumen.pub	MODERATE
China (later periods)	1:7 to 1:8	Dokumen.pub	MODERATE
Europe (medieval)	1:10 to 1:15	Goldsilverstacks.com	MODERATE

The significant gold-silver ratio differential created arbitrage opportunities—silver was worth approximately **twice as much in China as in Europe**, driving precious metal flows eastward along the Silk Road.

3.3 Currency systems across khanates

Table 3.8: Khanate Currency Systems

Khanate	Primary Currencies	Key Features	Source	Confidence
Yuan (China)	Paper jiaochao, silver ding, copper cash	Sole paper legal tender; sophisticated system	Guan et al. (2024)	HIGH
Ilkhanate	Gold dinars, silver dirhams, copper	Paper money attempted 1294, failed within months	Wikipedia	HIGH
Golden Horde	Silver dang (dirham)	Weight declined from 1.00–1.25g to 0.40–0.50g	Podhorski.net	HIGH
Chagatai	Silver dirhams	Weights 1.0–2.2 grams depending on mint	Numista	HIGH

The Ilkhanate's paper money experiment under Gaykhatu Khan (September 12, 1294) represents a fascinating failure. The chao notes—direct imitations of Yuan currency with Chinese characters—caused bazaar riots in Tabriz and were withdrawn within months.

Table 3.9: Golden Horde Coinage Details

Metric	Value	Source	Confidence
Dang weight (early 13th c.)	1.00–1.25 grams	Podhorski.net	HIGH
Dang weight (late 13th c.)	0.40–0.50 grams	Podhorski.net	HIGH
Dang weight (1310–1360)	1.45–1.60 grams (stabilized)	Podhorski.net	HIGH
Copper pul exchange	16 puls = 1 dannik	Podhorski.net	HIGH
Dinar:Dirham ratio	1:6	Podhorski.net	MODERATE
Major mints	Sarai, Qrim, Bulghar, Khwarizm, Kaffa	WorthPoint	HIGH

3.4 Interest rates and lending

Table 3.10: Interest Rate Data

Context	Rate	Source	Confidence
Ortaq lending rate	**100% compound annual**	Richmond Federal Reserve	HIGH
Government loans to ortaq	"Low interest" (unspecified)	Wikipedia; Columbia AFE	MODERATE

The 100% compound annual rate charged by ortaq merchants was enabled by the Mongols being "unencumbered by usury laws of Islamic and Christian worlds." This extraordinarily high rate explains ortaq merchants' poor reputation among Chinese populations.

3.5 Yuan fiscal deficits

Table 3.11: Revenue vs. Expenditure

Year	Revenue (1,000 ding)	Expenditure (1,000 ding)	Deficit	Source	Confidence
1292	2,979	3,638	-659	Yuanshi ch. 17, 22, 24	HIGH
1308	4,000	13,400	-9,400	Yuanshi ch. 22, 24	HIGH
1311	4,000	18,000	-14,000	Yuanshi ch. 22, 24	HIGH

The explosive growth in deficits—from 659,000 ding in 1292 to **14 million ding by 1311**—directly preceded the transition to fiat currency and foreshadowed the dynasty's eventual monetary collapse.

---

4. Institutional indicators

4.1 Great Khan succession (1206–1259)

Table 4.1: Great Khans of the Unified Empire

Khan	Reign Dates	Length	Accession Method	Death	Source	Confidence
Chinggis Khan	1206–1227	21 years	Kurultai proclamation	Natural (fell from horse)	Secret History	HIGH
Tolui (Regent)	1227–1229	~2 years	Regency (youngest son tradition)	Stepped aside	Britannica	HIGH
Ögedei Khan	1229–1241	12 years, 3 months	Kurultai election (designated heir)	Natural (alcoholism)	Rashid al-Din	HIGH
Töregene Khatun (Regent)	1241–1246	~5 years	Widow's regency	Power transferred	Britannica	HIGH
Güyük Khan	1246–1248	~2 years	Kurultai election	Died en route to confront Batu	Wikipedia	HIGH
Oghul Qaimish (Regent)	1248–1251	~3 years	Widow's regency	Overthrown, executed	Britannica	HIGH
Möngke Khan	1251–1259	8 years, 1 month	Kurultai (organized by Sorghaghtani and Batu)	Dysentery (siege of Chongqing)	Wikipedia	HIGH

Great Khan Aggregate Statistics (1206–1259):

- Total Great Khans (excluding regents): **4**
- Average reign length: **10.75 years**
- Violent succession rate: **0%** (all natural deaths or illness)
- Regencies: **3** (Tolui, Töregene, Oghul Qaimish)

4.2 Khanate rulers post-1260 fragmentation

Table 4.2: Yuan Dynasty Rulers (1260–1368)

#	Khan/Emperor	Temple Name	Reign	Length	Succession	Death
1	Kublai Khan	Shizu	1260–1294	34 yrs	Civil war victory	Natural (78)
2	Temür Khan	Chengzong	1294–1307	13 yrs	Designated heir	Natural (41)
3	Külüg Khan	Wuzong	1307–1311	4 yrs	Succession struggle	Natural (29)
4	Ayurbarwada	Renzong	1311–1320	9 yrs	Brother's abdication	Natural (34)
5	Shidebala	Yingzong	1320–1323	3 yrs	Hereditary	**Assassinated**
6	Yesün Temür	—	1323–1328	5 yrs	Coup	Natural (34)
7	Ragibagh	—	1328	1 month	Hereditary	**Murdered** (8)
8	Tugh Temür	Wenzong	1328–1329	4 months	Coup	Abdicated
9	Qoshila	Mingzong	1329	6 months	Brother's abdication	**Murdered**
10	Tugh Temür (2nd)	Wenzong	1329–1332	3 yrs	Fratricide	Natural (28)
11	Irinjibal	Ningzong	1332	2 months	Hereditary	Natural (6)
12	Toghon Temür	Huizong	1333–1368	35 yrs	Designated	Fled to Mongolia

Yuan Dynasty Statistics:

- Total rulers: **12**
- Average reign: **9 years**
- Violent successions: **4** (Shidebala, Ragibagh, Qoshila, plus Yesün Temür via coup)

- **Violent succession rate: 33%**

Table 4.3: Khanate Succession Statistics Summary

Khanate	Period	Total Rulers	Avg Reign	Violent Succession %
Great Khans	1206–1259	4	10.75 years	0%
Yuan Dynasty	1260–1368	12	9 years	33%
Ilkhanate	1256–1335	9	8.8 years	44%
Golden Horde	1227–1359	11	12 years	36%
Chagatai Khanate	1227–1363	~12	8 years	42%

The contrast between the **0% violent succession** under the Great Khans and the **33–44%** rates in successor khanates illustrates the stability costs of imperial fragmentation.

4.3 Kurultai assemblies

Table 4.4: Major Kurultai Events

Date	Location	Major Decisions	Source	Confidence
1206	Source of Onon River	Temüjin proclaimed Chinggis Khan	Secret History	HIGH
1211	—	Decision to invade Jin China	Multiple	HIGH
1229	Kode'e Aral (Kelüren River)	Ögedei elected Great Khan	Britannica	HIGH
1235	Karakorum	Western campaign (Europe) ordered	Multiple	HIGH
1246	Near Karakorum	Güyük elected; Papal envoy Carpine present	Carpine; Britannica	HIGH
1251	Kode'e Aral	Möngke elected; power shift Ögedeid→Toluid; 77–300 opponents purged	Jami' al-tawarikh	HIGH

The kurultai system provided legitimacy through collective decision-making: chiefs debated, voted, and decisions were considered divinely sanctioned by Tengri. Absence from a kurultai signaled disloyalty.

4.4 Yasa law code

The yasa (jasagh) presents methodological challenges—it was primarily oral law, kept secret, with only fragmentary reconstruction possible from multiple sources.

Table 4.5: Yasa Establishment

Metric	Value	Source	Confidence
Date of establishment	c. 1206 (kurultai proclamation)	Wikipedia; Jami' al-tawarikh	MODERATE
Chief judge	Shigi Qutuqu (Chinggis's adopted son)	Secret History	HIGH
Enforcement overseer	Chagatai Khan (second son)	Juvayni; Wikipedia	HIGH
Format	Oral; possibly recorded in "blue-script book"	David Morgan	HIGH
Duration of use	Through Yuan dynasty to 1368	Cambridge Core	HIGH

Table 4.6: Key Yasa Provisions (Reconstructed)

Category	Provision	Penalty	Source
Religious	Complete religious freedom	Tax exemption for clergy	Genghis Khan decree
Military	Division into tens, hundreds, thousands, ten thousands	—	Juvayni
Military	Soldiers receive arms from officer, maintain readiness	Death for defection	Secret History
Criminal	Adultery	Death	Juvayni; Rashid al-Din
Criminal	Theft (repeat offense)	Death	Rashid al-Din
Criminal	False witness	Death	Makrizi
Political	No khan proclaimed without kurultai	Death	Juvayni
Diplomatic	Ambassadors have immunity	Protected	Juvayni

4.5 Yam postal system

The yam (Persian/Russian) or örtöö (Mongolian) postal network represents the Mongol Empire's most quantifiable institutional achievement.

Table 4.7: Yam System Infrastructure (Yuan China)

Component	Quantity	Source	Confidence
Postal stations in China	>1,400	Marco Polo; Wikipedia	HIGH
Total stations empire-wide	~10,000	Marco Polo estimate	MODERATE
Horses maintained	~50,000	Wikipedia; Marco Polo	HIGH
Oxen	1,400	Wikipedia	HIGH
Mules/donkeys	6,700–8,500	Wikipedia	MODERATE
Boats	6,000	Wikipedia	MODERATE
Carts	400	Wikipedia	MODERATE
Camels	5,700	Wikipedia	MODERATE
Dogs (sled transport)	200–7,000	Sources vary	MODERATE

Table 4.8: Yam Operational Specifications

Metric	Value	Source	Confidence
Station spacing	20–40 miles	Multiple	HIGH
Foot-runner posts	Every 3 miles	Marco Polo	MODERATE
Horses per station	15–400 (varied by importance)	Ghazan reforms; Marco Polo	MODERATE
Daily courier distance	200–300 km (120–190 miles)	Wikipedia	HIGH
Speed improvement	10x faster than previous systems	World History Encyclopedia	MODERATE
Ilkhanate stations (1251)	119 relay stations (northern sectors)	Grokipedia	MODERATE

4.6 Religious tolerance policies

Table 4.9: Religious Tax Exemptions

Religion	Mongol Term	Exemptions Granted	Duration	Confidence
Buddhism	Toyin	Taxation, labor, public service	1206–1368	HIGH
Christianity (Nestorian)	Erke'üd	Taxation, labor, public service	1206–1368	HIGH
Taoism	Xiansheng	Taxation, labor, public service	1206–1368	HIGH
Islam	Dashmad	Taxation, labor, public service	1206–1368	HIGH
Russian Orthodox	—	Taxation; special protection	13th–15th c.	HIGH

The institutional basis in the yasa was explicit: "May the Buddhists, Christians, Taoists and Muslims be exempt from all taxation and may they pray to God and continue offering us blessings" (Christopher Atwood). Religious tolerance was pragmatic governance—reducing resistance and facilitating administration across diverse populations.

---

5. Military indicators

5.1 Tumen organization

The decimal military system—arban (10) → zuun (100) → mingghan (1,000) → tumen (10,000)—provided the organizational backbone of Mongol armies.

Table 5.1: Military Unit Structure

Unit	Mongolian Term	Nominal Size	Typical Composition	Confidence
Squad	Arban	10 men	Base unit	HIGH
Company	Zuun/Jaghun	100 men	~60 archers, 40 lancers	HIGH
Regiment	Mingghan	1,000 men	~5,000 mounts	HIGH
Division	Tumen	10,000 men	~50,000 mounts	HIGH
Army Corps	Ordu	30,000–100,000	Campaign force	MODERATE

Table 5.2: Total Tumen Counts

Date	Event	Count	Theoretical Strength	Source	Confidence
1206	Unification	95 mingghan	95,000	Secret History (NamuWiki)	MODERATE
1227	Chinggis's death	—	129,000	Secret History (Quora)	MODERATE
c.1250	Möngke period	~155 tumen combined	~1,550,000 theoretical	NamuWiki	LOW

Critical Note: Timothy May observes tumens "usually only maintained about **60% of authorized population**"—actual operational strength of a "10,000-man" tumen was typically **~6,000 men**.

5.2 Major campaigns with army sizes

Table 5.3: Khwarazmian Campaign (1219–1221)

Metric	Value	Source	Confidence
Dates	1219–1221	Wikipedia	HIGH
Commander	Chinggis Khan	Wikipedia	HIGH
Mongol army size	75,000–200,000 (most: 90,000–150,000)	Multiple	MODERATE
Khwarazmian forces	200,000–400,000 (likely exaggerated)	Battle of Parwan	LOW
Sverdrup estimate (Tolui's Khorasan force)	~7,000 men	Wikipedia	MODERATE
Duration	~2 years	—	HIGH

Table 5.4: Jin Dynasty Campaign (1211–1234)

Metric	Value	Source	Confidence
Initial invasion	1211	Wikipedia	HIGH
Capture of Zhongdu	1215	Wikipedia	HIGH
Final conquest	1234	Wikipedia	HIGH
Duration	23 years	—	HIGH
Mongol forces (initial)	30,000–100,000	Quora; NamuWiki	MODERATE

Table 5.5: Baghdad/Ilkhanate Campaign (1256–1258)

Metric	Value	Source	Confidence
Commander	Hulagu Khan	Wikipedia	HIGH
Mongol army size	100,000–150,000 (~120,000–138,000)	Multiple	MODERATE-HIGH
Chinese siege engineers	~1,000 technicians; 3,000 giant crossbows	Multiple	MODERATE
Baghdad garrison	~30,000 troops	Wikipedia	MODERATE
"One in ten" conscription claim	Yes (from entire empire)	War History Online	MODERATE

Table 5.6: European Campaign (1237–1242)

Metric	Value	Source	Confidence
Commanders	Batu Khan, Subutai (operational)	Wikipedia	HIGH
Total invasion force	40,000–130,000	Britannica; NWE	MODERATE
Force at Hungary	40,000–80,000	Multiple	MODERATE
Force at Poland (diversionary)	10,000–20,000 (1–2 tumen)	Wikipedia	MODERATE

Table 5.7: Japan Invasion Attempts

Campaign	Date	Fleet Size	Troop Strength	Losses	Confidence
First Invasion	1274	800–900 ships	23,000–40,000	~13,500 did not return	MODERATE
Second Invasion	1281	4,400–5,000 ships	100,000–140,000	Catastrophic (typhoon)	MODERATE

Thomas Conlan (2001) argues these numbers are "inflated by a factor of ten or more"—a significant revisionist position.

5.3 Siege casualties: the methodological problem

Medieval siege casualty figures are among history's most unreliable data points. Chronicles routinely reported deaths exceeding plausible city populations.

Table 5.8: Siege Casualties (Comparative)

City	Date	Medieval Reports	Modern Estimates	Scholarly Consensus
Baghdad	1258	800,000–2 million	90,000–200,000	MODERATE confidence for lower range
Merv	1221	700,000–1.3 million	Unquantifiable	Reports logistically impossible
Nishapur	1221	1–2.4 million	Unquantifiable	City had <200,000 population
Kiev	1240	Near-total	~38,000–48,000 (survivors: 2,000)	MODERATE

Baghdad Scholarly Positions:

Scholar/Source	Estimate	Notes
Hulagu Khan (contemporary)	~200,000	His own estimate
Martin Sicker	~90,000	Conservative modern
Ian Frazier (New Yorker)	200,000–1,000,000	Range cited
Hamdallah Mustawfi (1330)	800,000	Persian historian

5.4 Battle casualties

Table 5.9: Battle of Mohi/Sajó River (April 11, 1241)

Force	Size Estimate	Casualties	Source	Confidence
Hungarian	10,000–100,000 (Sverdrup: ~10,000)	10,000–60,000+ killed	Multiple	MODERATE
Mongol	15,000–80,000 (Sverdrup: ~15,000)	"Several hundred to several thousand"	Wikipedia	LOW

Aftermath: ~25% of Hungarian population killed in invasion; "nearly half of inhabited places destroyed."

Table 5.10: Battle of Ain Jalut (September 3, 1260)

Force	Size Estimate	Casualties	Confidence
Mamluk	~20,000	Unknown	MODERATE
Mongol	10,000–20,000 (one tumen plus vassals)	"Almost the whole army" destroyed	MODERATE
Mongol commander Kitbuqa	—	Killed in battle	HIGH

5.5 Carl Sverdrup's military analysis

Sverdrup's *The Mongol Conquests* (2017) represents the most significant recent revision of Mongol military history.

Table 5.11: Sverdrup's Revisions

Topic	Sverdrup's Finding	Traditional View	Confidence
Battle of Mohi armies	Hungarians: ~10,000; Mongols: ~15,000	100,000 vs. 80,000	MODERATE
Tolui's Khorasan force	~7,000 men	Higher estimates	MODERATE
Khwarazmian total army	~40,000	200,000–400,000	MODERATE
Bukhara garrison	2,000–5,000	Juvaini: 20,000	MODERATE

Sverdrup's Key Conclusions:

1. Mongols often **outnumbered** opponents in battle (contrary to popular belief)
2. Success came from **tactics and strategy**, not terror or numbers
3. Mongols suffered **several defeats and setbacks**

4. Terror was **not** a deliberate strategy
5. Medieval chronicler numbers are **grossly exaggerated**

---

6. Data gaps and source quality assessment

6.1 Critical data gaps by category

Table 6.1: Data Gap Summary

Category	Gap Description	Impact	Remediation Potential
Economic	Silk Road trade volumes (quantified)	Cannot assess commercial scale	LOW—no contemporary records
Economic	Ortaq profit-sharing percentages	Cannot model partnership economics	LOW—varied by contract
Economic	Non-Yuan khanate fiscal data	Incomplete comparative picture	MODERATE—Persian archives
Demographic	Pre-conquest population baselines	Undermines mortality calculations	LOW—inherent limitation
Demographic	Accurate siege casualties	Historical debates unresolvable	LOW—propagandistic sources
Financial	Chagatai/Golden Horde detailed finances	Incomplete monetary history	MODERATE—numismatic evidence
Military	Actual operational tumen strength	True army sizes uncertain	MODERATE—archaeological evidence

6.2 Source quality hierarchy

Table 6.2: Source Reliability Assessment

Source Type	Example	Reliability	Notes
Contemporary administrative records	Yuan Shi fiscal data (1328)	HIGH	Direct taxation records
Contemporary eyewitness accounts	Giovanni da Pian del Carpine (Kiev 1246)	HIGH	On-site observation
Contemporary memoir	Marco Polo (with caveats)	MODERATE	Exaggeration concerns
Contemporary chronicles	Juvayni, Rashid al-Din	MODERATE	Political bias; number inflation
Medieval Persian histories	Hamdallah Mustawfi	MODERATE	Written decades later
Modern scholarly reconstructions	Maddison GDP estimates	LOW-MODERATE	Methodologically contested
Popular synthesis	McEvedy & Jones mortality figures	LOW	Now considered weak methodology

---

7. Cross-empire comparison table

This table completes the CAS Framework Empire Knowledge Base seven-empire comparative analysis.

Table 7.1: Seven-Empire Quantitative Comparison

Metric	Roman	Byzantine	Tang	Abbasid	Ottoman	Venetian	Mongol
Peak GDP/capita	$570–857	$680–770	~$600–800	~$600–700	~$600–800	$1,000–1,500	~$450–600
Duration	503 yrs	1,123 yrs	289 yrs	508 yrs	623 yrs	1,100 yrs	**162 yrs**
Total Rulers	69–77	94	20–21	37	36	~120	5 GK + 44 khanate
Violent Succession %	**62–75%**	~57%	25–30%	25–35%	40–50%	**<10%**	0% (GK); 33–44% (khanates)
Peak Territory	5.0M km²	5.0M km²	5.0M km²	11.1M km²	5.2M km²	~45K km²	**24M km²**
Currency Stability	LOW	**HIGH**	MODERATE	MOD-HIGH	MODERATE	**HIGH**	LOW
Avg. Reign Length	6.5–7.3 yrs	~12 yrs	~14 yrs	~14 yrs	~17.3 yrs	~9 yrs	10.75 yrs (GK)

Table 7.2: Key Comparative Findings

Finding	Empire	Value	Significance
Largest territory	**Mongol**	24 million km²	Largest contiguous land empire in history
Longest duration	Byzantine	1,123 years	Most durable complex state
Most violent succession	Roman	62–75%	Most dangerous leadership position
Most stable succession	Venetian	<10%	Electoral system prevented violence
Most stable currency	Byzantine	700+ years	Solidus as "dollar of Middle Ages"
Shortest duration	**Mongol**	162 years	Rapid expansion, rapid fragmentation
Highest GDP/capita	Venetian	$1,000–1,500	Commercial republic wealth

Confidence Levels for Comparison Data:

- Peak territory: HIGH (well-documented)

- Duration: HIGH (political dates clear)
- GDP/capita: LOW-MODERATE (pre-modern GDP inherently uncertain)
- Violent succession: HIGH for Roman, Byzantine, Venetian; MODERATE for others
- Currency stability: HIGH (numismatic evidence strong)

---

8. Methodological notes and scholarly disagreements

8.1 Key scholarly debates

Table 8.1: Major Contested Questions

Topic	Position A	Position B	Current Consensus
Total conquest mortality	40–60 million (McEvedy & Jones)	10–15 million (revisionists)	Leaning toward lower estimates
Mongol army sizes	Traditional: 100,000+ per campaign	Sverdrup: typically 30,000–50,000	Sverdrup gaining acceptance
Yuan census accuracy	Reflects real population decline	Severely underregistered	Underregistration likely
Yuan paper money cause of failure	Over-issuance	Loss of silver backing	Both factors; transition to fiat crucial
Strategic terror	Deliberate policy	Opportunistic/circumstantial	May: used "judiciously"

8.2 Source chronology issues

Table 8.2: Contemporary vs. Later Sources

Source	Date Composed	Events Described	Time Gap	Reliability Impact
Secret History of the Mongols	c. 1240	1162–1227	0–78 years	HIGH for later events
Juvayni	c. 1260	1219–1260	0–40 years	MODERATE
Rashid al-Din	c. 1310	1206–1310	0–100 years	MODERATE (used archives)
Yuan Shi	1370	1206–1368	2–164 years	HIGH for fiscal; MODERATE for events
Hamdallah Mustawfi	1330	1219–1330	0–111 years	MODERATE
Marco Polo	c. 1298	1271–1295	3–27 years	MODERATE (exaggeration concerns)

8.3 Nomadic economy measurement challenges

Traditional GDP metrics—designed for sedentary agricultural societies—struggle to capture Mongol economic activity:

- **Wealth storage**: Livestock, textiles, precious metals rather than land
- **Value creation**: Trade facilitation, protection services, tribute extraction
- **Production**: Pastoral nomadism defies output measurement
- **Taxation basis**: Herds, commerce, conquest rather than agricultural surplus

The Yuan dynasty portion provides conventional fiscal data, but the nomadic core economy remains fundamentally unquantifiable by standard metrics.

---

9. Conclusions and synthesis

9.1 Quantitative profile of the Mongol Empire

The Mongol Empire presents a distinctive quantitative profile: **maximum territorial extent** (24 million km²) combined with **minimum duration** (162 years) among the seven empires in this comparative framework. This expansion-to-fragmentation ratio—approximately **148,000 km² gained per year of existence**—is unmatched in world history.

Table 9.1: Summary Metrics

Domain	Key Finding	Confidence
Peak population ruled	~100 million (25% of world)	MODERATE
Peak territory	24 million km² (16% of land)	HIGH
Duration (unified)	53 years (1206–1259)	HIGH
Duration (with Yuan)	162 years (1206–1368)	HIGH
Great Khan violent succession	0%	HIGH
Post-fragmentation violent succession	33–44%	HIGH
Yuan central revenue (1328)	27.7 million silver taels	HIGH
Salt tax share	46% of Yuan revenue	HIGH
Yam stations	>1,400 (China); ~10,000 (empire)	HIGH/MODERATE
Horses in yam system	~50,000	MODERATE

9.2 Comparative position in seven-empire framework

The Mongol Empire occupies extremes in several comparative dimensions:

- **Largest** territory ever ruled from a single center
- **Shortest** duration among the seven empires
- **Most stable** Great Khan succession (0% violent) but rapid destabilization after fragmentation
- **Most sophisticated** paper money system (Yuan jiaochao) but also most dramatic monetary collapse
- **Most extensive** communications infrastructure (yam postal network)
- **Lowest** measurable GDP/capita (reflecting nomadic economy measurement challenges)

9.3 Data quality assessment

Table 9.2: Overall Confidence by Category

Category	Overall Confidence	Best Data	Worst Data
Economic	MODERATE	Yuan 1328 fiscal records	Trade volumes
Demographic	LOW-MODERATE	Chinese censuses	Siege casualties
Financial	HIGH (Yuan); LOW (others)	Paper money issuance	Khanate currencies
Institutional	HIGH	Ruler lists; yam statistics	Yasa specifics
Military	MODERATE	Unit organization	Actual army sizes

The Mongol Empire's quantitative data quality is fundamentally asymmetric: Yuan dynasty Chinese records provide excellent fiscal, monetary, and administrative data, while the nomadic heartland and non-Chinese khanates remain poorly documented. Conquest mortality figures should be treated as indicators of catastrophic events rather than reliable demographic statistics.

9.4 Implications for CAS Framework analysis

The Mongol case offers several insights for Complex Adaptive Systems analysis of empires:

1. **Rapid expansion creates fragmentation pressure**: The empire's 53-year unified period suggests scaling limits for steppe-based governance systems
2. **Institutional innovation has limits**: The yam system and paper money represented advanced institutional solutions, but could not prevent political fragmentation or monetary collapse
3. **Succession stability is conditional**: The 0% violent succession rate under Great Khans contrasts sharply with 33–44% in successor states, suggesting institutionalized succession (kurultai) worked only under specific conditions
4. **Economic measurement reflects governance type**: The sharp contrast between quantifiable Yuan fiscal data and unquantifiable nomadic economy metrics illustrates how state capacity shapes historical visibility

This appendix completes the Infonomics Project's seven-empire quantitative data retrofit workstream, providing comparable metrics across Roman, Byzantine, Tang, Abbasid, Ottoman, Venetian, and Mongol empires for CAS Framework analysis.

Primary sources referenced

- **Secret History of the Mongols** (c. 1240) — Mongol perspective on Chinggis Khan's life

- **Jami' al-tawarikh** (Rashid al-Din, c. 1310) — Persian comprehensive history
- **Tarikh-i Jahangushay** (Juvayni, c. 1260) — Persian chronicle of Mongol conquests
- **Yuan Shi** (1370) — Chinese dynastic history
- **Jin Shi** — Jin dynasty records
- **Song Shi** — Song dynasty records
- **Giovanni da Pian del Carpine** (1247) — Papal envoy eyewitness account
- **Marco Polo** (c. 1298) — Venetian merchant memoir
- **Yuandianzhang** — Yuan legal/administrative collection

Secondary sources referenced

- Abu-Lughod, Janet L. *Before European Hegemony: The World System A.D. 1250–1350* (1989)
- Allsen, Thomas T. *Commodity and Exchange in the Mongol Empire* (1997)
- Durand, John. *The Population Statistics of China, A.D. 2–1953* (1960)
- Enkhbold, Enerelt. "The role of the ortoq in the Mongol Empire" *Central Asian Survey* (2019)
- Guan, Palma & Wu. "The Rise and Fall of Paper Money in Yuan China" *Economic History Review* (2024)
- Guan, Ma & Zhai. "State Capacity in Imperial China" Peking University Working Paper (2024)
- May, Timothy. *The Mongol Art of War* (2007)
- McEvedy, Colin & Richard Jones. *Atlas of World Population History* (1978)
- Morgan, David. *The Mongols* (1990)
- Rossabi, Morris. *Khubilai Khan: His Life and Times* (1988)
- Sverdrup, Carl. *The Mongol Conquests* (2017)
- von Glahn, Richard. *Fountain of Fortune: Money and Monetary Policy in China* (1996)